Individual Assignment 6 & 7

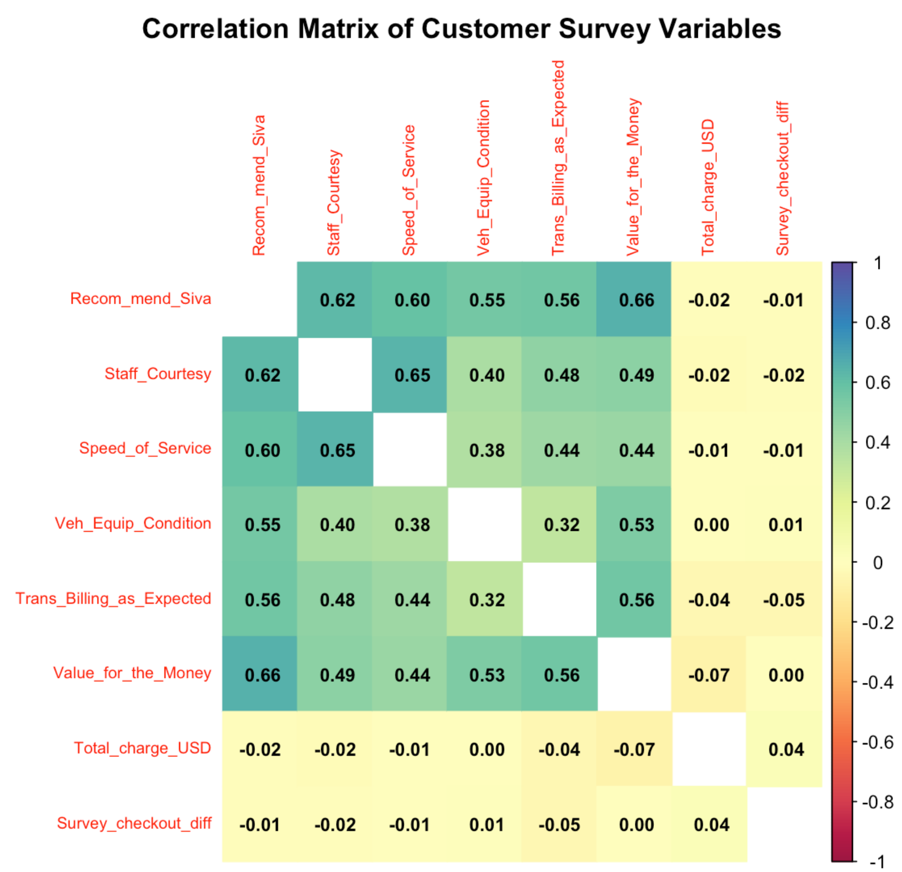
Chintan S. Jikkar

[U01989702]

```
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(corrplot)

## corrplot 0.95 loaded

library(factoextra)

## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa

library(RColorBrewer)

siva <- read.csv("/Users/chintanjikkar/Desktop/PACE/Fall 25/Visual Analytics/Data/siva.csv", na = c("NA", "
"))
str(siva)

## 'data.frame':    53815 obs. of  29 variables:
##  $ xgra_n1clb_nbr           : int  51407 23460 53417 14382 40539 53945 35983 43669 29279 14254 ...
##  $ Siva_Rental_Number       : int  67041 56084 70279 15105 49797 71102 43104 54673 33940 14950 ...
```

```
##  $ rent_area_loc           : int  156 204 181 1515 259 165 177 276 2167 953 ...
##  $ Date_of_Survey          : chr  "5/18/2011" "2/5/2011" "6/14/2011" "1/4/2010" ...
##  $ Day_of_Week             : chr  "Wednesday" "Saturday" "Tuesday" "Monday" ...
##  $ Time                    : chr  "7:48:30" "22:06:37" "5:35:48" "23:58:56" ...
##  $ Survey_Type             : chr  "SV Web Sol." "SV Web Sol." "SV Web Sol." "SV Web Sol." ...
##  $ Purpose_of_Rental       : chr  "Bus." "Leis. / Pers." "Bus." "Leis. / Pers." ...
##  $ Recom_mend_Siva         : int  8 8 8 7 9 9 9 6 9 5 ...
##  $ Staff_Courtesy          : int  9 8 7 8 9 9 9 9 8 7 ...
##  $ Speed_of_Service        : int  8 8 8 7 9 9 9 8 9 5 ...
##  $ Veh_Equip_Condition     : int  9 5 8 8 9 9 9 9 8 8 ...
##  $ Trans_Billing_as_Expected: int  9 8 8 8 9 9 9 9 9 7 ...
##  $ Value_for_the_Money     : int  9 7 8 8 9 9 9 6 7 7 ...
##  $ Area                    : chr  "01602 - LOVE FIELD AP TX" "07286 - VALLEJO CA OAP" "01850 - RICHMOND
VA AP" "05743 - PICO CA OAP" ...
##  $ loc_nm                  : chr  "DALLAS LOVE FIELD" "VALLEJO HLE" "RICHMOND INTL AP" "PICO HLE" ...
##  $ ga_region_desc          : chr  "SOUTHWEST REGION" "WESTERN REGION" "MID ATLANTIC REGION" "WESTERN RE
GION" ...
##  $ xgra_ckot_ts            : chr  "5/15/2011" "1/31/2011" "6/12/2011" "12/23/2009" ...
##  $ xgra_ckin_ts            : chr  "5/17/2011" "2/3/2011" "6/13/2011" "1/3/2010" ...
##  $ xgra_vclass_reserv      : chr  "C" "A" "F" "D" ...
##  $ xgra_veh_class          : chr  "Q4" "B" "YF" "YF" ...
##  $ rent_loc_type           : chr  "AP" "OFF AP" "AP" "OFF AP" ...
##  $ cust_tier_code          : chr  "FG" "N1" "RG" "RG" ...
##  $ booking_channel_code    : chr  "SIVA.COM" "SIVA.COM" "SIVA.COM" "SIVA.COM" ...
##  $ col34_total_charges     : num  247.3 128 75.8 468.5 42.8 ...
##  $ col38_currency          : chr  "USD" "USD" "USD" "USD" ...
##  $ Total_charge_USD        : num  247.3 128 75.8 468.5 42.8 ...
##  $ Survey_checkout_diff    : int  2 3 2 2 2 4 2 2 6 2 ...
##  $ booking_channel_dummy   : int  1 1 1 1 1 0 0 0 1 1 ...

numerical_vars <- siva %>%
  select(
    Recom_mend_Siva,
    Staff_Courtesy,
    Speed_of_Service,
    Veh_Equip_Condition,
```

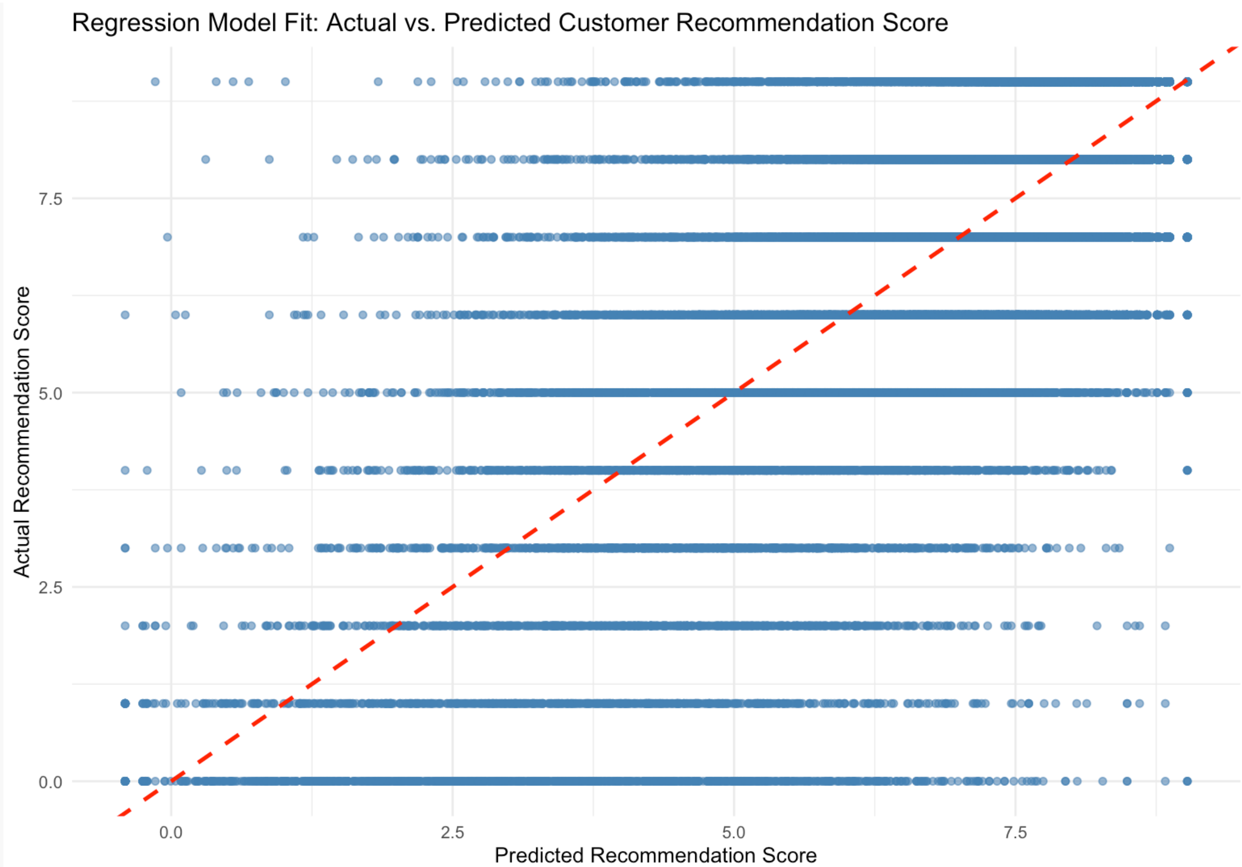
```r
    Trans_Billing_as_Expected,
    Value_for_the_Money,
    Total_charge_USD,
    Survey_checkout_diff
  ) %>%
  mutate(across(everything(), as.numeric))

siva_clean <- na.omit(numerical_vars)

correlation_matrix <- cor(siva_clean)
corr_colors <- colorRampPalette(RColorBrewer::brewer.pal(11, "Spectral"))(200)

corrplot(
  correlation_matrix,
  method = "color",
  col = corr_colors,
  addCoef.col = "black",
  tl.cex = 0.7,
  number.cex = 0.8,
  diag = FALSE,
  title = "Correlation Matrix of Customer Survey Variables",
  mar = c(0, 0, 2, 0)
)
```




```r
regression_model <- lm(
  Recom_mend_Siva ~ Staff_Courtesy + Speed_of_Service +
    Veh_Equip_Condition + Trans_Billing_as_Expected +
    Value_for_the_Money,
  data = siva_clean
)

summary(regression_model)
```

```
##
## Call:
## lm(formula = Recom_mend_Siva ~ Staff_Courtesy + Speed_of_Service +
##     Veh_Equip_Condition + Trans_Billing_as_Expected + Value_for_the_Money,
##     data = siva_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.0267 -0.5159  0.1299  0.6015  9.1416
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -0.409053   0.029629  -13.81   <2e-16 ***
## Staff_Courtesy             0.248569   0.004678   53.13   <2e-16 ***
## Speed_of_Service           0.195483   0.003446   56.72   <2e-16 ***
## Veh_Equip_Condition        0.180413   0.002935   61.48   <2e-16 ***
## Trans_Billing_as_Expected  0.156543   0.003327   47.05   <2e-16 ***
## Value_for_the_Money        0.267408   0.003629   73.69   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.28 on 52340 degrees of freedom
## Multiple R-squared:  0.6247, Adjusted R-squared:  0.6247
## F-statistic: 1.743e+04 on 5 and 52340 DF,  p-value: < 2.2e-16
```

```r
siva_clean$Predicted_Recom <- predict(regression_model)

ggplot(siva_clean, aes(x = Predicted_Recom, y = Recom_mend_Siva)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed", color = "red", linewidth = 1) +
  labs(
    title = "Regression Model Fit: Actual vs. Predicted Customer Recommendation Score",
    x = "Predicted Recommendation Score",
    y = "Actual Recommendation Score"
  ) +
  theme_minimal()
```




```
k_value <- 3
cluster_data <- siva %>%
  select(Staff_Courtesy, Speed_of_Service, Veh_Equip_Condition,
         Trans_Billing_as_Expected, Value_for_the_Money) %>%
  na.omit()

scaled_data <- scale(cluster_data)
```

```r
set.seed(2101)
kmeans_model <- kmeans(scaled_data, centers = k_value, nstart = 25)

fviz_cluster(
  kmeans_model,
  data = as.data.frame(scaled_data),
  ellipse.type = "convex",
  geom = "point",
  palette = c("#FF6F61", "#FFE082", "#4DB6AC"),
  ggtheme = theme_minimal(),
  main = paste("Cluster Visualization using Principal Components (k =", k_value, ")")
)
```

Cluster Visualization using Principal Components (k = 3)
5.0
2.5
0.0
-2.5
Dim2 (14.9%)
0
2
4
6
8
Dim1 (57.6%)
cluster
1
2
3